MULTIPLE SCLEROSIS AND OTHER DEMYELINATING CENTRAL NERVOUS SYSTEM DISORDERS*

IS-NPMSD FALL 2025

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(*ADAPTED FROM WORK OF CATHERINE WEST MD, DRPH)

LEARNING OBJECTIVES:
STUDENTS WILL BE ABLE TO:

1. Recognize a ‘typical’ clinical presentation of MS
2. Recall environmental and genetic features that appear to be causally related to MS
3. Describe the pathophysiology and pathogenesis of MS and recognize associated pathology
4. State the major differences between MS and Guillain-Barre Syndrome

LEARNING OBJECTIVES

THE STUDENT WILL BE ABLE TO:

5. List common neurologic and ancillary signs and symptoms of MS
6. Describe the 3 patterns of the clinical course of MS and identify the most common of the 3.
7. Generate a differential Dx for MS and outline an appropriate work up, including understanding of the importance of MRI
8. Describe difference between treatments for acute relapses and disease modifying therapy and identify when each is used
9. Identify common acute relapse treatments and disease modifying agents for MS
    - Describe proposed mechanism of action of dalfampridine and importance of testing for JC virus

LEARNING OBJECTIVES

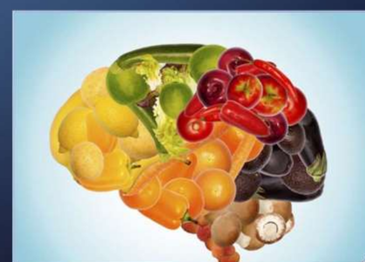
STUDENTS WILL BE ABLE TO:

10. List common treatments for ancillary symptoms
11. Describe current opinions regarding pregnancy in women with MS
12. Describe basic features of ADEM
13. Recognize structural issues affecting treatment of MS particularly those in marginalized communities.
14. Recognize osteopathic principles related to treatment of MS including the relationship of nutrition on structure and function and body's ability to self regulate and heal.

FYI:

- Watch for the **"Boards" Bolt**

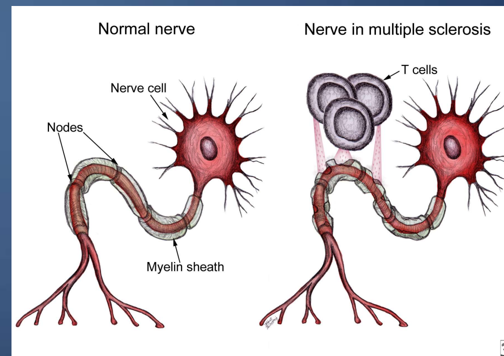
- **Bonus** session discussing principles of functional nutrition related to neurodegenerative disorders, including MS



MULTIPLE SCLEROSIS

- Most common chronic neurological disorder seen in younger people.
- Inflammatory immune disorder mediated by auto-reactive T-lymphocytes
  - Lead to demyelination, gliosis and axon loss in the CNS
- Can be recurrent episodic or progressive symptoms (or combination) reflected by dysfunction of CNS, Brainstem and/or Spinal Cord

- Most common in middle aged adults
  - 15-50 y/o typical age range
- Female > Male
- Cause of significant morbidity (partially due to younger age of onset and progressive nature over time)
- Management will include disease modification and symptom management.
  - Newer DMT's have improved the prognosis;
  - Nutrition and lifestyle modifications can have significant effect yet often overlooked.

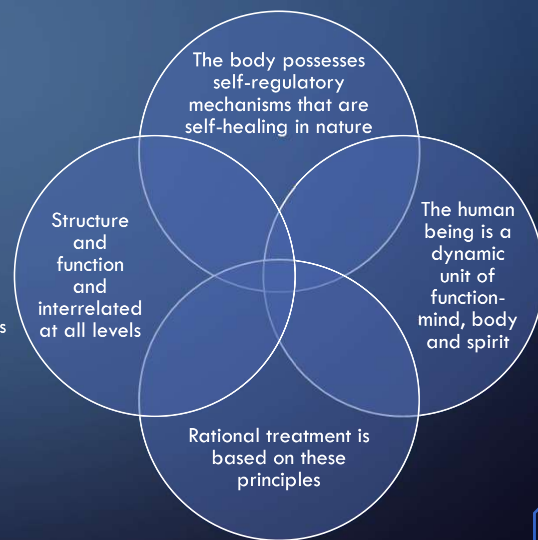


OSTEOPATHIC PRINCIPLES

Person is Dynamic Unit of Mind, Body and Spirit

- What structural issues are affecting mind, body and spirit.
- ACE's' Chronic Stress, Minority Stress
- Environmental Factors/pollution; Food security (swamps, deserts, apartheid)
- 1868 Charcot reported link between stress and MS

- Structure and Function are Reciprocally Inter-related at all levels
  - What is the structure of the mitochondria? How can we enhance that?
  - Somatoskeletal Structure: OMM to improve flexibility, myofascial (contains immune cells), reduce pain
- Body has Innate Ability to Heal Itself
- Rational Treatment Based on these Principles



KT is a 28-year-old female with PMHX significant for acute onset vision loss at age 22 (currently resolved), intermittent numbness in her face (resolved) who presents today with CC of 'dizzy/vertigo' over the last two days. She notes the hot weather seems to be an aggravating factor.

She denies headache, trauma or other neurological symptoms. She denies fever, rash or any known infections. She denies any recent travel. Prior vision loss and facial numbness were on left side with no known eliciting or alleviating factors noted; both resolved with in 3 months of onset;

Fam Hx is non-contributory

Soc Hx is + for tobacco use; She has worked on a farm for >10 years but has not had health insurance benefits.

Common Manifestations

- Younger, female
- Vison loss/Optic Neuritis
- Sensory Abnormalities very common
- Vertigo
- Heat Sensitivity
- Fatigue
- Remitting and relapsing symptoms
- Clinically distinct episodes with partial or total remission in between
- Smoking, CVD, Obesity are risk factors
- Lack of adequate health care access/cost

Obj. 1: Recognize a 'typical' clinical presentation of MS

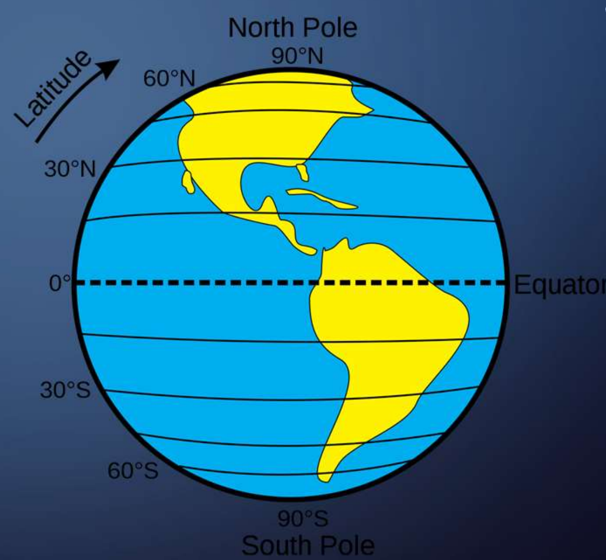
Obj 1

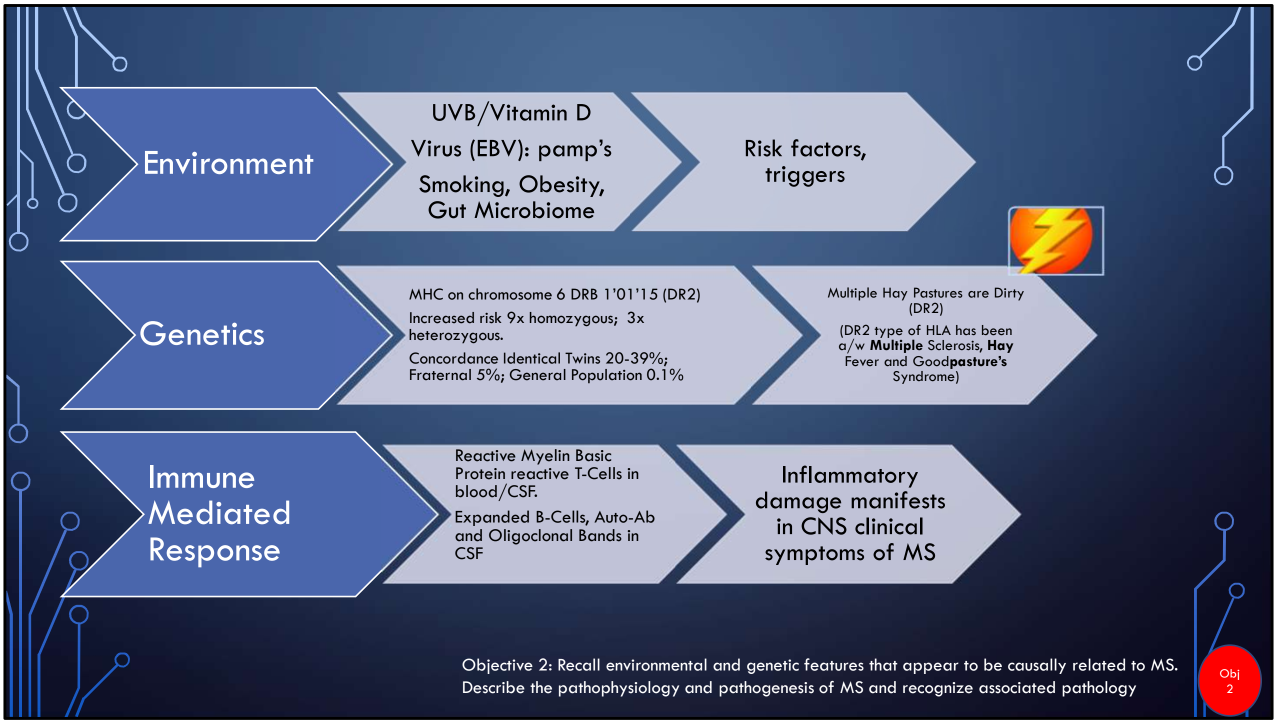
AGE/GENDER

- Peak incidence 20 to 30 years
  - 70% between 21 and 40 years
- Female>Male 2.3:1
  - Female: 2-3x as likely as Male to develop MS

EPIDEMIOLOGY

- Increased risk in northern latitudes
  - Patients moving to northern hemisphere before puberty have same risk as being born there; Moving after puberty, risk as those in Southern hemisphere
  - Possibly related to environmental factors (virus exposure?)
  - Vitamin D Exposure (less sun in northern hemispheres
- Possible connection to viral exposure (EBV, Lyme)





HLA-DRB 15:01: 3x risk; Antigen presenting cells to CD4 Cells; Interferon B, targets activated CD4; glatiramer mimics antigen presentation (over 200 genes; innate and humoral); S1PR agonists help maintina immune cells in lymphoid tissue; teriflunomide, alters immune cell physiology; (antigen presenting cells, activate CD4 T cells >stimulate B cells) HLA-DRB a/w ability to present both myelin and EBV (molecular mimictry) CYP24A1 inactivates Vitmain D
HLA: human leukocyte antigen (MHC cell surface receptor) produced by chromosome 6; Of DR type; ligand for T-Cells. Multiple (sclerosis) Hay (fever) goodPastuer's syndrome are DiRty (DR2)

OBJECTIVE 3:
DESCRIBE THE PATHOPHYSIOLOGY AND PATHOGENESIS OF MS AND RECOGNIZE ASSOCIATED PATHOLOGY

Inflammation:

Disruption of Blood Brain Barrier

Inflammatory T-Cells and Macrophages

B-Cell auto-antibody to Myelin Sheath

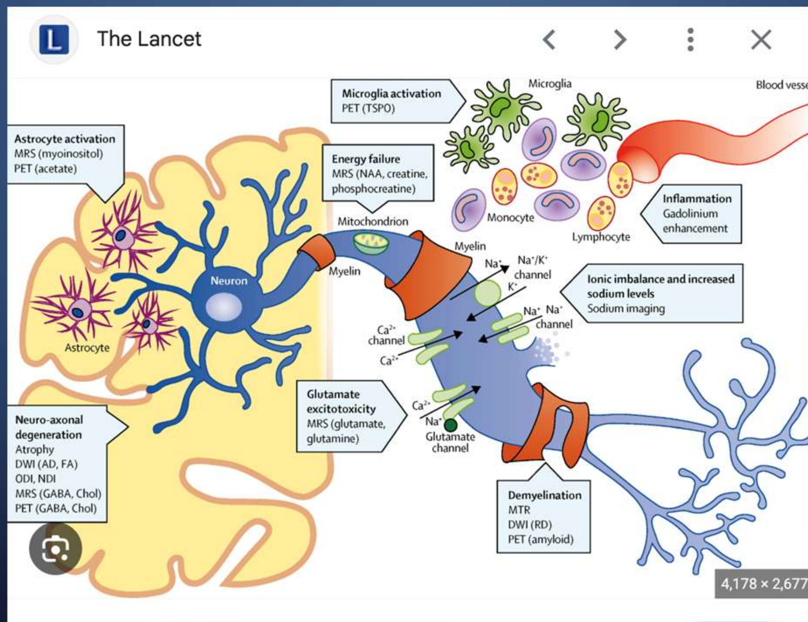
Some Remyelination

Partial or Total Axonal Destruction

Dendritic Loss

Gliosis

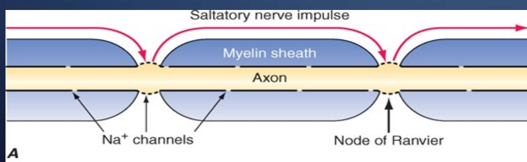
As with nearly all chronic disease, inflammation is a driving cause. What diet and lifestyle modifications can reduce inflammation?


The Lancet
Microglia activation
PET (TSPO)
Microglia
Blood vessel
Astrocyte activation
MRS (myoinositol)
PET (acetate)
Energy failure
MRS (NAA, creatine, phosphocreatine)
Mitochondrion
Monocyte
Lymphocyte
Inflammation
Gadolinium enhancement
Myelin
Neuron
Myelin
Na^+
Na^+/K^+ channel
K^+
Ionic imbalance and increased sodium levels
Sodium imaging
Na^+ Na^+ channel
Ca^{2+} channel
Ca^{2+}
Astrocyte
Glutamate excitotoxicity
MRS (glutamate, glutamine)
Ca^{2+}
Na^+
Glutamate channel
Neuro-axonal degeneration
Atrophy
DWI (AD, FA)
ODI, NDI
MRS (GABA, Chol)
PET (GABA, Chol)
Demyelination
MTR
DWI (RD)
PET (amyloid)
4,178 × 2,677

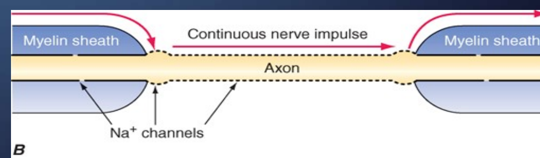
FYI

OBJECTIVE 3:
DESCRIBE THE PATHOPHYSIOLOGY AND PATHOGENESIS OF MS AND RECOGNIZE ASSOCIATED PATHOLOGY

- DESTRUCTION AND LOSS OF MYELIN SHEATH, LEADS TO 'BARE' AXON.
- EXPOSED VOLTAGE DEPENDENT K CHANNELS
- RELATIVE LACK OF VOLATGE DEPENDENT NA CHANNELS
- CONDUCTION BLOCK ACROSS AXON
- NA CHANNELS EVENTUALLY IMPROVE/RECOVER BUT IS NOT NORMAL CONDUCTION
- TEMPERATURE AND METABOLIC DISTRUBANCES MAY AFFECT CONDUCTION
- K CHANNEL BLOCKERS MAY BE THERAPEUTIC TREATMENT OPTION

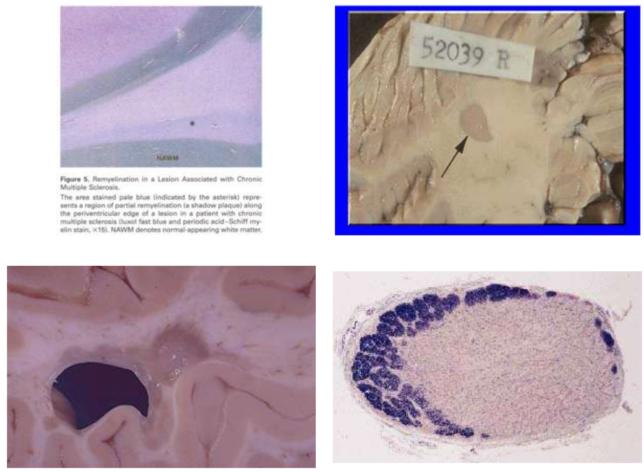


Source: Longo DL, Fauci AS, Kasper DL, Hauser SL, Jameson JL, Loscalzo J: *Harrison's Principles of Internal Medicine, 18th Edition*: www.accessmedicine.com



Source: Longo DL, Fauci AS, Kasper DL, Hauser SL, Jameson JL, Loscalzo J: *Harrison's Principles of Internal Medicine, 18th Edition*: www.accessmedicine.com

IMAGE FROM HARRISON'S ONLINE



John H Noseworthy, Claudia Lucchinetti, Moses Rodriguez, Brian G Weinshenker. Multiple sclerosis. N Engl J Med. 2000 Sep 28;343(13):938-52.

Could diet/lifestyle influence myelin production? Neuron Health?

- Myelin contains high amounts of cholesterol and essential fatty acids, especially DHA
  - Fatty Fish, Venison
  - Body can make but need adequate co-factors, energy; alcohol inhibitor of EPA/DHA pathways.
- Zinc may help prevent myelin breakdown; Necessary for immune function.
- Research indicates oligodendrocytes may repair and divide during sleep.

MS IS A "WHITE MATTER" DISEASE OF THE CNS THAT IS DISSEMINATED IN SPACE (MULTIPLE LESIONS IN CHARACTERISTIC LOCATIONS) AND TIME (DEVELOPMENT OF NEW LESIONS OVER TIME)

DISSEMINATION IN SPACE (LOCATION)

- BRAIN, EYE, SPINAL CORD
- SYMPTOMS NOT EXPLAINED BY SINGLE LESION;
- MULTIPLE LESIONS ON MRI.
- CIS: CLINICALLY ISOLATED SYNDROME (INITIAL PRESNTATION)

DISSEMINATION IN TIME (DIT)

- ADDITIONAL LESIONS APPEAR SUBSEQUENTLY
- MAY BE ASSOCIATED WITH CLINCIAL SIGNS/SYMPTOMS
- MAY ONLY BE SEEN ON IMAGING (RADIOLOGIC ISOLATED SYNDROME)

- MAY BE DRAMATIC OR MILD ONSET OF SYMPTOMS
- PAROXYSMS: SUDDEN/BRIEF (SECONDS TO MINUTES) OF DYSFUNCTION

Obj 5. List common neurologic and ancillary signs and symptoms of MS

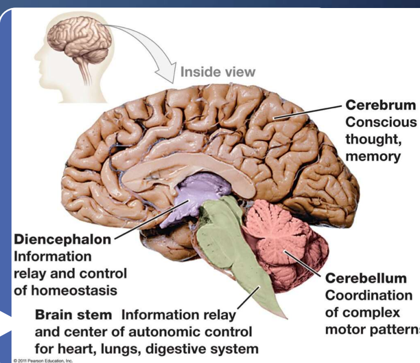
CLINCIAL PRESENTATION DEPENDS ON LOCATION AND ANATOMOICAL PATHWAYS

LOCATION

- CORTICAL
- COLOSSAL
- PERIVENTRICULAR
- DEEP WHITE MATTER
- BRAIN STEM
- CEREBELLUM
- SPINAL CORD

ANATOMY

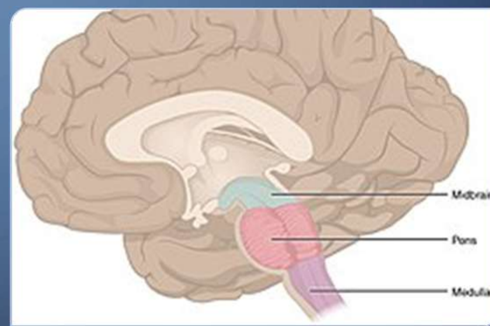
- PATHWAYS
- DECUSSATIONS
- CONTINGUOUS STRUCTURES



BRAINSTEM:

- Sensory and Motor connections pass thru brainstem
  - Corticospinal and Corticobulbar Tract (motor)
  - Dorsal Column Medial Lemniscus (fine touch/vibration)
  - Spinothalamic Tract/Anterolateral System: pain/temperature/itch

- Nuclei of Cranial Nerves (except CNI/II) in brainstem
  - Diplopia, vertigo, dysphagia, dysarthria
- “Relay” station



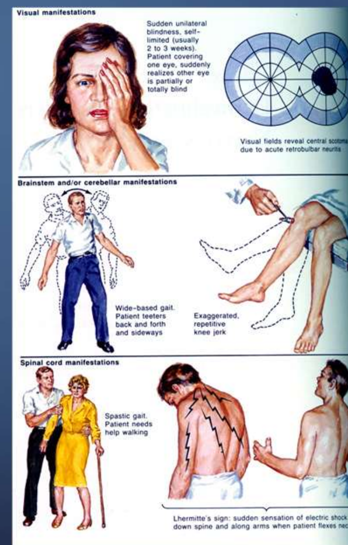
NEURO EXAM

- Mental Status
- Cranial Nerves
- Motor
- Coordination and Gait
- Sensory
- Reflexes

MENTAL STATUS

COGNITIVE/MOOD

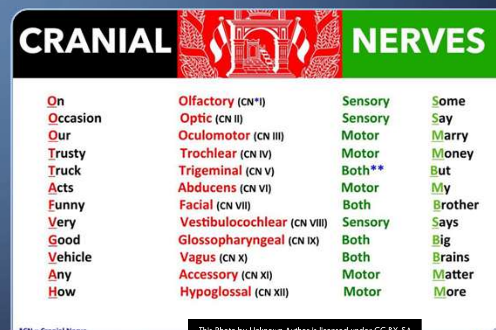
- MENTAL FATIGUE
- POOR MEMORY
- POOR ATTENTION
- DEPRESSION
- MOOD/AFFECT DISSOCIATION



Obj 5. List common neurologic and ancillary signs and symptoms of MS

CRANIAL NERVES

- CN1: Olfactory
- CNII: Optic Nerve; Check acuity, fields, fundoscopy (swollen disk or pallor) optic neuritis is common, afferent pupillary defect
- CNIII: Oculomotor Nerve; Pupils reactivity; Extra-ocular muscles, nystagmus; Intranuclear ophthalmoplegia via disruption of MLF.
- CNIV/CNVI: Extra-ocular muscles, trochlear and lateral rectus;
- CNV: Trigeminal; Sensation loss/pain on face; Jaw Jerk
- CNVII: Facial Nerve: Strength of facial muscles, weakness.
- CNVIII: Vestibulocochlear; vertigo
- CNIX: Glossopharyngeal
- CNX: Vagus;
- CNXI: Spinal Accessory Nerve;
- CNXII: Hypoglossal: protrude tongue; mild dysarthria; scanning speech, loss of melody.



CRANIAL NERVES

On	Olfactory (CN*I)	Sensory	Some
Occasion	Optic (CN II)	Sensory	Say
Our	Oculomotor (CN III)	Motor	Marry
Trusty	Trochlear (CN IV)	Motor	Money
Truck	Trigeminal (CN V)	Both**	But
Acts	Abducens (CN VI)	Motor	My
Funny	Facial (CN VII)	Both	Brother
Very	Vestibulocochlear (CN VIII)	Sensory	Says
Good	Glossopharyngeal (CN IX)	Both	Big
Vehicle	Vagus (CN X)	Both	Brains
Any	Accessory (CN XI)	Motor	Matter
How	Hypoglossal (CN XII)	Motor	More

OBJ. 5

CRANIAL NERVES:

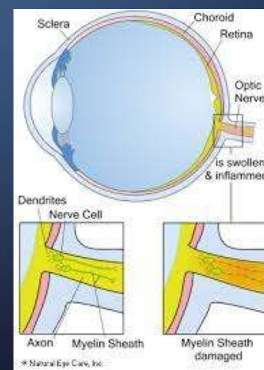
VISUAL: OPTIC NEURITIS IS COMMON

- SUDDEN BLURRED OR LOSS OF VISION
- LOSS OF COLOR VISION
- UNILATERAL PAIN BEHIND ON EYE (COMMON, 90%)
- VISUAL FIELD DEFECTS

Subjective/Symptoms: May be CC by patient.

ASK as part of history: trouble w/vision, hearing? Dizzy/light headed/spinning/unsteady;

Problems swallowing, difficulties with speech?

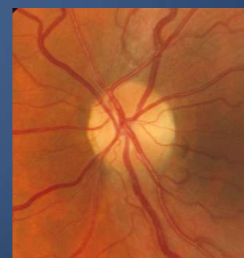


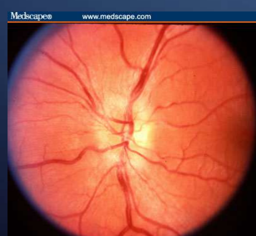
OBJ. 5

OPTIC NEURITIS: FUNDOSCOPIC EXAM

Normal optic disc

In acute optic neuritis, disc may be normal, pale or swollen.





This is a swollen disc—occurs when demyelination has occurred anterior to lamina cribrosa (about 1/3 of cases)

Pale optic disc
Optic neuritis that has resolved with residual optic nerve atrophy

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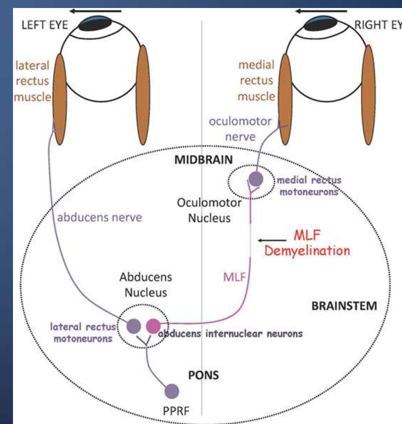
http://www.medscape.org/viewarticle/571660_2

http://oph.ucsd.edu/residents/learning/opticdisc/chapter4/4_17.htm

SIGNS: OBJECTIVE

CRANIAL NERVES/BRAINSTEM

- INO: INTRANUCLEAR OPTHALMOPLEGIA
  - Board review often use: INO: IPISILATERAL ADDUCTION FAILURE, NYSTAGMUS OPPOSITE (problem in MLF, medial longitudinal fasciculus)
  - Although nystagmus is very common with INO, it is not absolutely necessary. The actual opthalmoplegia occurs due to the defect in the MLF.
  - MS is likely the most common of INO, but other damage to MLF can cause it as well.
- Afferent Pupillary Defect:
- Scanning speech, ataxic speech with loss of melody
  - https://www.youtube.com/watch?v=2MscEYj1LSg
  - https://youtu.be/dIXSa79kVkA



OBJ. 5


INTERNUCLEAR
OPHTHALMOPLEGIA
(INO)

INTRA NUCLEAR OPTHALMOPLEGIA

Ipsilateral ADDuction deficit

Nystagmus Opposite (commonly)

MOTOR:

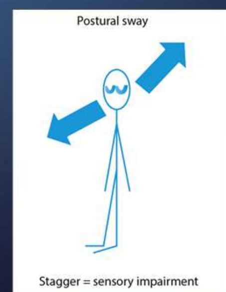
- Movement: assess gait, may be disordered/ataxic; poor coordination; slow. Severity Scales often related to ambulatory function.
- Muscle Bulk: muscle wasting
- Tone: increased tone, spasticity; cramping
- Strength: weakness common; check pronator drift
- Rapid Alternating Movements (tests basal ganglia and cerebellum function)

- Bladder/Sexual Dysfunction common. Don't forget to ask in your history/ROS.

OBJ. 5

SENSATION:

- Light Touch
- Temperature
- Vibration
- Proprioception
- Romberg (sensory ataxia)



Disturbances in sensation one of most common manifestations of MS. May occur in one or more limbs, face or sensory levels;

Patient may complain of numbness, tingling.

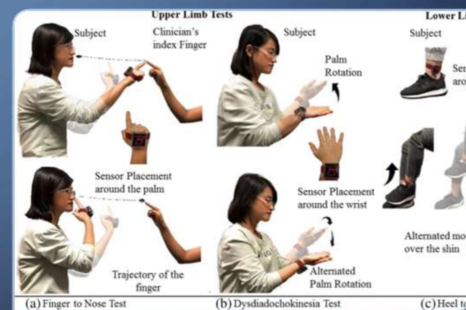
Pain more frequent with lesions in the Spinal Cord.

OBJ. 5

Spinothalamic: LST: temperature; VST: crude touch; pressure; Dorsal Columns: vibration, 2-pt discrimination, proprioception; Spinocerebllar: unconscious proprioception.

COORDINATION AND GAIT

- Finger to Nose Test: may see dysmetria with intention tremor; overshoot finger to nose
- Heel to Shin
- Cerebellum can be affected in MS leading to symptoms such as ataxia, dysmetria, dysarthria, scanning speech, nystagmus and tremor. (you might notice some of these in other parts of your neuro exam such as CN testing or gait).
- Difficulties with fine motor skills such as buttoning a shirt or handwriting may occur. Tremor is common.



OBJ. 5

COORDINATION AND GAIT

ASSESSMENT OF GAIT IS A VALUABLE ASSESSMENT TOOL, PARTICULARLY IN MS.

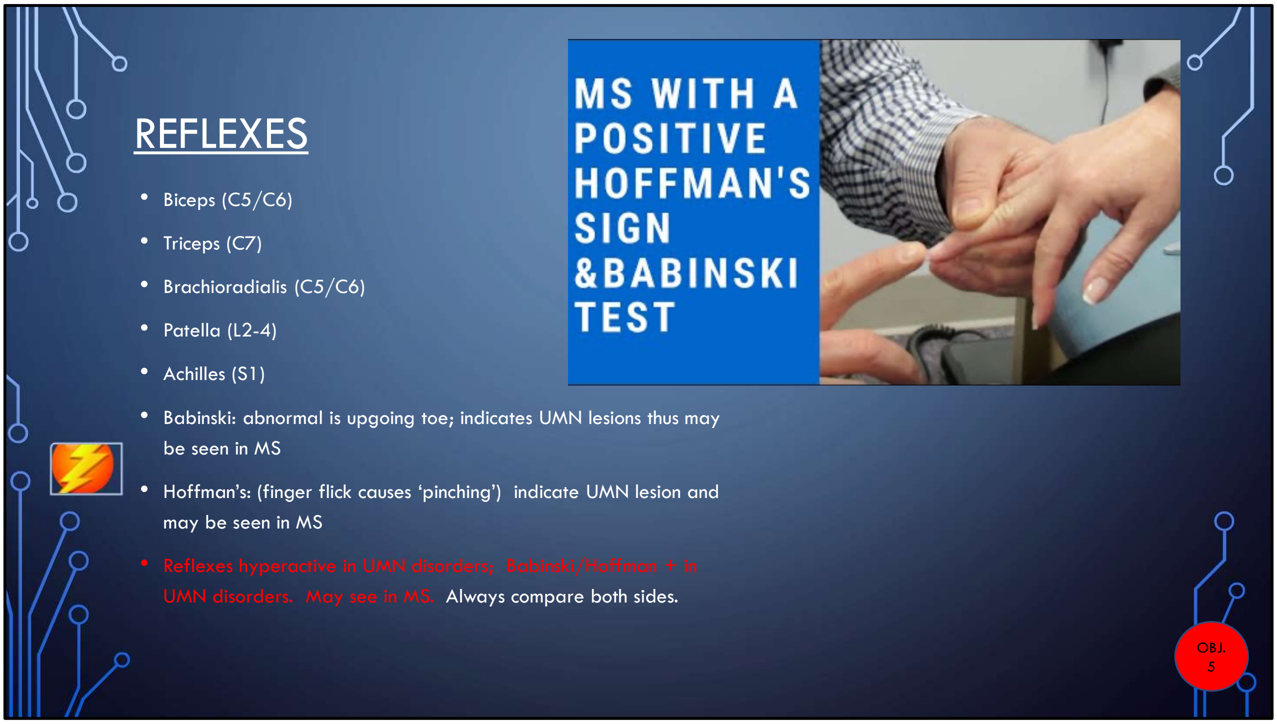
- Posture
- Stride Length: diminished
- Stance (wide: cerebellar, proprioception; narrow: spasticity)
- Pace: diminished speed
- Arm Swing
- Heel to Toe Walk (assess strength)

Ataxic gait often seen in MS, due to cerebellar dysfunction and corticospinal lesions with weakness. Sensory disturbances may also contribute to gait problems. Diminished QOL and Falls are a major risk factor.

Standardized timed and patient reported measures (25 minute timed walk, 6 minute walk, MS Walking Scale) can help monitor gait and indicate progression or improvements.

Treatments include OMT, physical therapy, graduated exercise, appropriate assistive devices and safe homes. DMT to reduce relapses, antispasmodic medications, diet, dalfampridine (K channel blocker)

OBJ. 5


REFLEXES
Biceps (C5/C6)
Triceps (C7)
Brachioradialis (C5/C6)
Patella (L2-4)
Achilles (S1)
Babinski: abnormal is upgoing toe; indicates UMN lesions thus may be seen in MS
Hoffman's: (finger flick causes 'pinching') indicate UMN lesion and may be seen in MS
Reflexes hyperactive in UMN disorders; Babinski/Hoffman + in UMN disorders. May see in MS. Always compare both sides.
MS WITH A POSITIVE HOFFMAN'S SIGN &BABINSKI TEST
OBJ. 5

“FAMOUS” SIGNS

- Lhermitte’s: electrical sensations down back and into limbs when neck is flexed (dorsal columns) (not specific)
- Uhthoff's Phenomenon: temporary worsening of symptoms exacerbated by heat.
  - Typically visual: blurred, diplopia
  - May be motor/sensory
  - Heat sensitivity manifested as fatigue
- Pulfrich’s pendulum: objects appear to move toward and away from viewer, like 3D movie

http://www.opticalillusion.net/optical-illusions/a-partly-fake-demo-of-pulfrichs-pendulum

Features typically suggestive of MS	Features typically NOT suggestive of MS
Relapse and Remissions	Steady Progression
Onset between age 15-50	Onset before 15 or after 50
Optic Neuritis	Cortical deficits such as aphasia, apraxia, neglect
Lhermitte's Sign	Rigidity, sustained dystonia
Internuclear Ophthalmoplegia (INO)	Convulsions
Fatigue	Early Dementia
Uhthoff Phenomena	Deficit developing with in minutes

OBJECTIVE 4: STATE THE MAJOR DIFFERENCES BETWEEN MS AND GUILLAIN-BARRE SYNDROME

MS IS ONE OF A GROUP OF DISORDERD CHARACTERIZED PRIMARILY BY DEMYELINATION

	MULTIPLE SCLEROSIS	**GUILLAINE-BARRE SYDNROME**
ABBREVIATION	MS	GBS OR AIDP
OTHER NAMES	La sclérose en plaque	Acute Inflammatory Demyelinating Polyneuropathy
LOCATION	**CNS**	**PNS**
CLINICAL	DISSEMINATED IN TIME AND SPACE	RAPIDLY ASCENDING PARALYSIS
KEY FINDINGS	VARIES; UMN PLUS OTHERS	AREFLEXIA
OTHER Demyelinating Disorders	Acute disseminated encephalomyelitis (ADEM) Acute Hemorrhagic Leukoencephalopathy Ropper, Samuels and Klein, Adam's and Victor's Principles of Neurology, Chapter 35	•Chronic Inflammatory Demyelinating Polyneuropathy (CIDP) SEE LECTURES ON PNS

ACUTE DISSEMINATED ENCEPHALOMYELITIS (ADEM)

- **"Multifocal inflammation and demyelination typically after infections or vaccination.**
  - **Chicken Pox, Measles Disease; MMR Vaccine; Flu/H1N1/HPV; Tdap**
    - 5% of cases follow vaccination (1-2 per 1,000,000 live measles vaccinations)
- **Presents with rapidly progressive multifocal neurologic symptoms, altered mental status"** (from USMLE or Boards)

ACUTE DISSEMINATED ENCEPHALOMYELITIS (ADEM)

- Rare; children>>adults
- Widely scattered small foci of periventricular inflammation and demyelination (NOT PLAQUES AS IN MS)
- Autoantibodies to myelin basic protein in CSF of some patients
- Abrupt onset, rapid progression late in course of viral illness
- Reappearance of fever, plus HA, meningismus, altered mental status and neurological signs common
- Rx is high-dose glucocorticoids;
  - Newer data indicate prognosis is better than previously indicated. However, it can be fatal in rare instances, and a small percentage have long term disability.
  - https://www.medscape.com/answers/1147044-104084/what-is-the-prognosis-of-acute-disseminated-encephalomyelitis-adem

Obj 12

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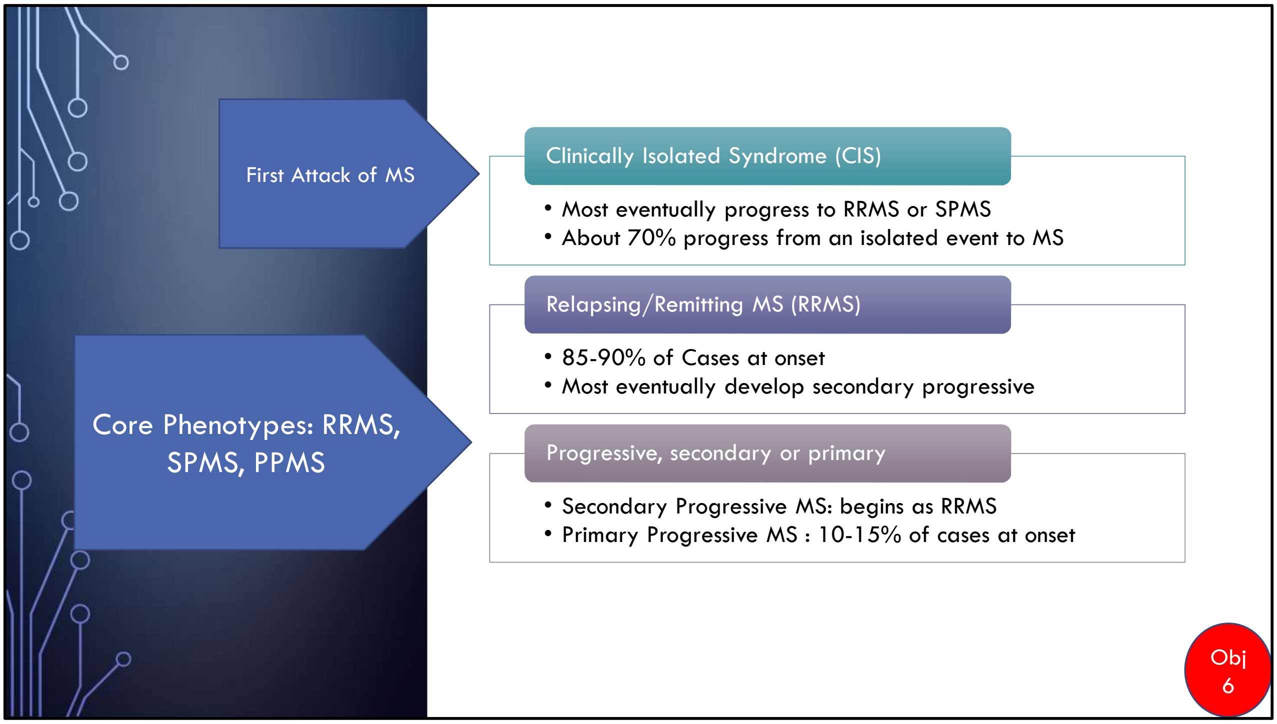
Classification of Multiple Sclerosis

Core Phenotypes:

1. Clinically Isolated Syndrome
2. Relapsing-Remitting: characterized time by symptoms/signs that remit followed by episodes of relapses.
3. Progressive: characterized by progression of s/s over time without episodes of remission. May be primary or secondary progressive

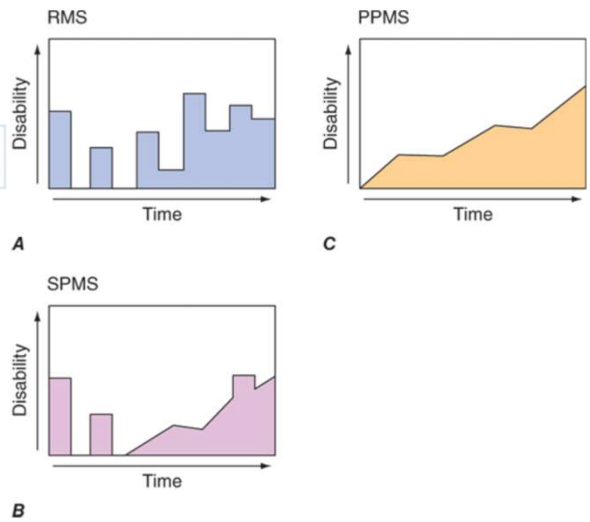
OBJ 6: DESCRIBE THE 3 PATTERNS OF THE CLINICAL COURSE OF MS AND IDENTIFY THE MOST COMMON OF THE 3.


First Attack of MS
Core Phenotypes: RRMS, SPMS, PPMS
Clinically Isolated Syndrome (CIS)
• Most eventually progress to RRMS or SPMS
• About 70% progress from an isolated event to MS
Relapsing/Remitting MS (RRMS)
• 85-90% of Cases at onset
• Most eventually develop secondary progressive
Progressive, secondary or primary
• Secondary Progressive MS: begins as RRMS
• Primary Progressive MS : 10-15% of cases at onset
Obj 6

A: Relapsing Remitting MS. 85-90% of cases at onset; most progress to SPMS over time.

C. Primary Progressive MS. 10-15% of cases at onset.

B. Secondary Progressive MS; always begins as RRMS.

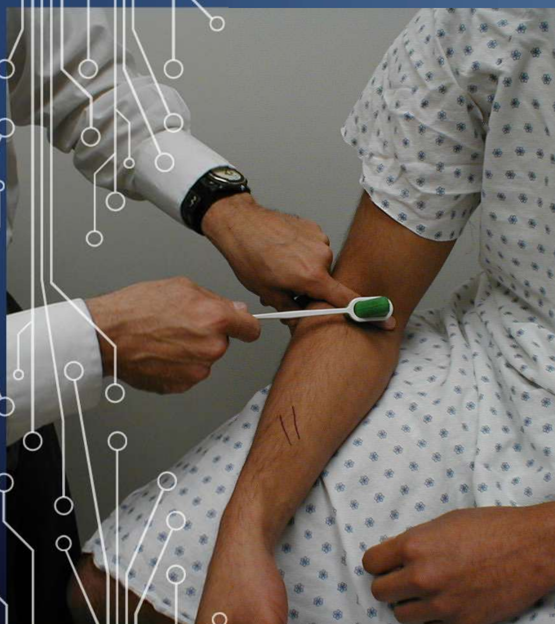


Source: J.L. Jameson, A.S. Fauci, D.L. Kasper, S.L. Hauser, D.L. Longo, J. Loscalzo: Harrison's Principles of Internal Medicine, 20th Edition

Obj 6

DIAGNOSIS

- Clinical and Diagnostic Assessment to Demonstrate:
  - Dissemination of lesions in space (DIS) and
  - Dissemination of lesions in time (DIT)
  - R/O other diagnosis



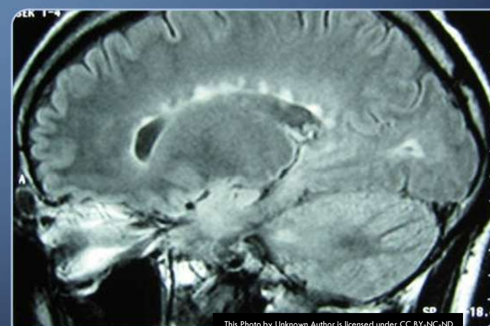
DIAGNOSIS AND WORK UP

Clinical Exam

- History: A thorough history including detailed neurological history is essential
- Physical: A detailed physical focusing on all 6 components of a neurological exam is essential
- Accurate Diagnosis, monitoring of progression and Drug Modifying Therapy all require initial and follow up detailed history and physical.
- Includes H&P details discussed above as well as use of clinical and patient generated monitoring scales. (Kurtze Disability Status Scale, Expanded Disability Status Scale, Patient Determined Disease Steps)

DIAGNOSTICS: MRI

- MRI with and without contrast
- If MRI not definitive then
  - Lumbar Puncture (LP)
  - Evoked Potentials (brainstem, visual, auditory, somatosensory)
- If suggestive of neuropathy or myelopathy then electromyogram (EMG) and nerve conduction studies (NCS)



Obj 7

MRI: >95% MS PATIENTS HAVE POSITIVE MRI FINDINGS; HOWEVER, >90% MRI FINDINGS ARE ASYMPTOMATIC

- **Gadolinium (Gd+) enhancement** T1-weighted images **(acute, lasts about 1 month, reflects breakdown of BBB)**
- **T2-weighted hyperintensity persists indefinitely** (evident at one month after acute phase began)
- Typical MS lesions:
  - Gd+ ring lesions
  - Oval/ovoid, well-circumscribed
  - T-2 lesions at least 3 mm in size (esp. important if > 6 mm)
  - Located in periventricular white matter (usually perpendicular to ventricular surface), corpus callosum, brainstem, cerebellum, or spinal cord

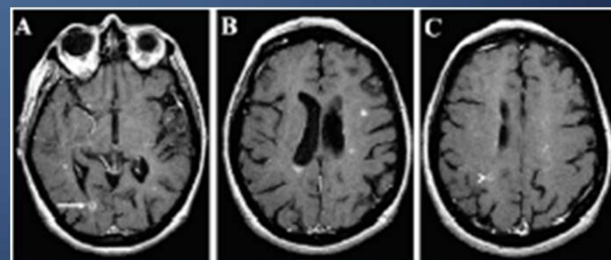


Fig. 4 Post-gadolinium T1-weighted spin-echo images of a 54 yo woman with relapsing remitting MS. Open ring lesions (A arrow, C arrowhead), typical of MS, as well as homogeneously enhancing lesions

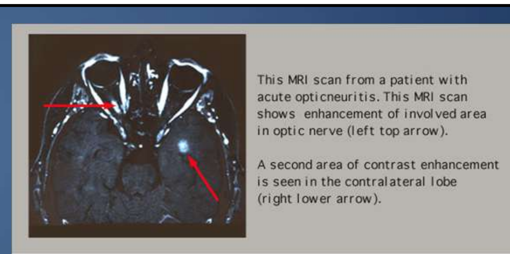
Neema M, Stankiewicz J, Arora A, Guss ZD, Bakshi R. MRI in multiple sclerosis: what's inside the toolbox? Neurotherapeutics. 2007 Oct;4(4):602-17

OBJ 7

MRI

- Cumulative Incidence (CI) of definite MS diagnosis over 15 years
  - Patient has symptom(s) of MS
    - If MRI is suggestive of MS; CI = 80%; **Start Disease-Modifying Treatment**
    - If MRI is normal; CI = 10-15%
      - Repeat MRI in 3 to 6 months; if suggestive of MS, **start Disease-Modifying Treatment**
      - If normal, no further MRI unless new symptoms
  - Patient has no MS symptoms
    - Incidental MRI findings suggestive of MS; CI = 25%

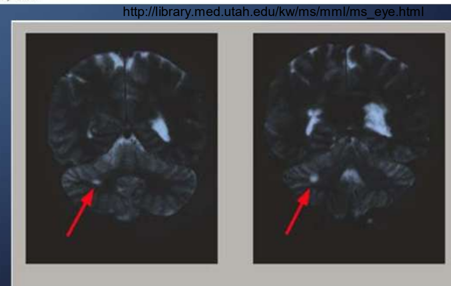
OBJ 7



Optic neuritis - MRI scan-KH

John Rose, M.D.

http://library.med.utah.edu/kw/ms/mml/ms_eye.html

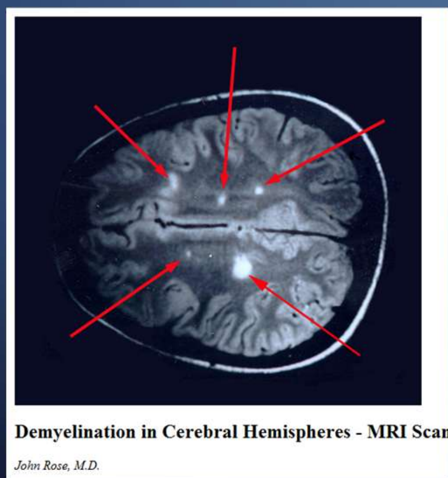


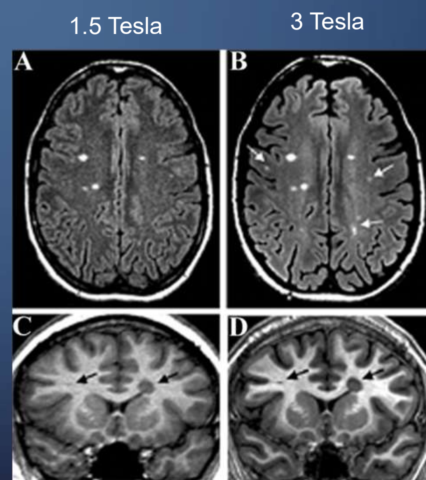
Area of Demyelination in Cerebellum - MRI scan

John Rose, M.D.

http://library.med.utah.edu/kw/ms/mml/ms_cerebellum.html

MAGNETIC RESONANCE IMAGING (MRI)





Neema M, Stankiewicz J, Arora A, Guss ZD, Bakshi R. MRI in multiple sclerosis: what's inside the toolbox? Neurotherapeutics. 2007 Oct;4(4):602-17.

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MAGNETIC RESONANCE IMAGING (MRI)

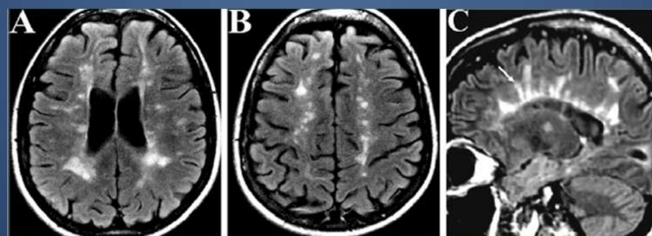

FIG. 1. 43 yo woman with relapsing–remitting MS (A–C). Note the typical hyperintense lesions, oval/ovoid in shape and seen in the periventricular white matter and juxtacortical gray-white junction. Several of the periventricular lesions directly contact the ventricular ependyma. In addition, most of the lesions are 5 mm or greater in diameter. The sagittal image (C) reveals "Dawson's fingers," characteristic of MS, believed to occur because of perivenular inflammation; the arrow indicates one of several "Dawson's fingers" seen on the sagittal image. (T-2 weighted images)

Dendritic loss and active gliosis

Neema M, Stankiewicz J, Arora A, Guss ZD, Bakshi R. MRI in multiple sclerosis: what's inside the toolbox? Neurotherapeutics. 2007 Oct;4(4):602-17.

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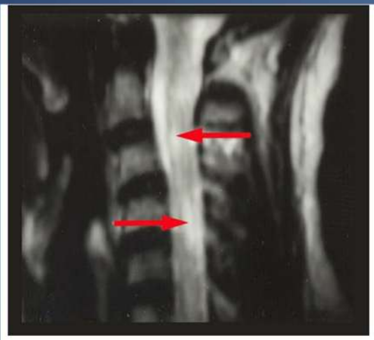
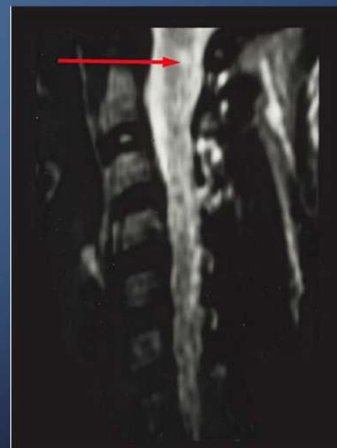
Obj 7

MAGNETIC RESONANCE IMAGING (MRI)



Demyelination in the Cervical Spinal Cord - MRI Scan

John Rose, M.D.



Demyelination in Cervical Spinal Cord - MRI Scan

John Rose, M.D.

http://library.med.utah.edu/kw/ms/mml/ms_cspine01.html

http://library.med.utah.edu/kw/ms/mml/ms_cspine02.html

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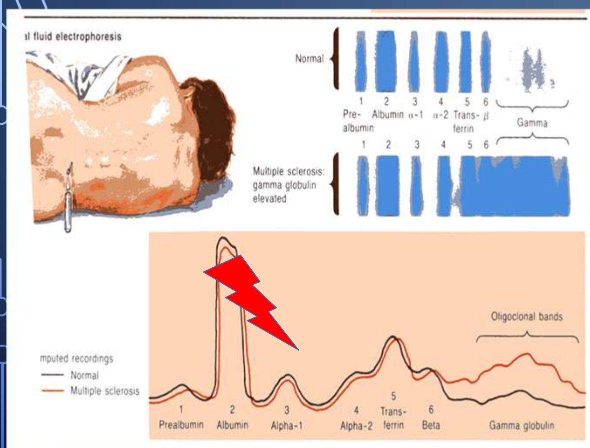
Obj 7

EXAMINATION OF CEREBROSPINAL FLUID (CSF)



INCREASED PROTEIN
- IGG
- OLIGOCLONAL BANDS ON ELECTROPHORESIS (75-95%)

MILD PLEOCYTOSIS

MYELIN METABOLITES
- MYELIN BASIC PROTEIN
- INCREASED IN CSF

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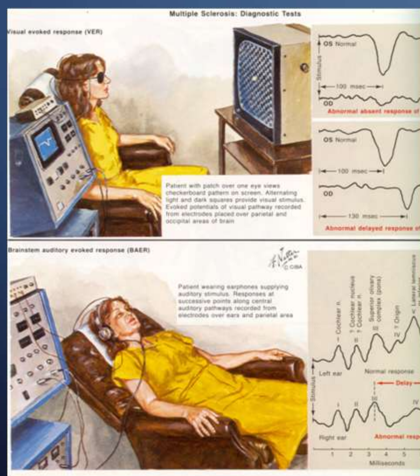
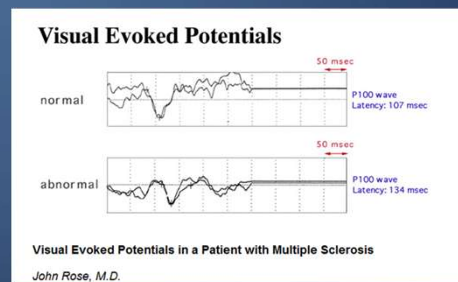
VISUAL AND BRAINSTEM EVOKED POTENTIALS





http://library.med.utah.edu/kw/ms/mml/ms_vep.html

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DIAGNOSIS: MCDONALD'S CRITERIA

	Number of lesions with objective clinical evidence	Additional data needed for diagnosis of MS
>/= 2 clinical attacks	a. >/= 2 b. 1 plus clear cut historical evidence of a previous attack involving lesion in a distinct anatomical location c. 1	a. None b. None c. Dissemination in space demonstrated by an additional clinical attack implicating a different CNS site or by MRI
1 clinical attack	a. >2 b. 1	a. Dissemination in time demonstrated by an additional clinical attack or by MRI; OR demonstration of CSF-specific oligoclonal bands b. Dissemination in space demonstrated by an additional clinical attack implicating a different CNS site or by MRI **AND** Dissemination in time demonstrated by additional clinical attack or by MRI **OR** CSF-specific oligoclonal bands.

Differential diagnosis of multiple sclerosis

Differential diagnosis of multiple sclerosis
Inflammatory disease
Acute disseminated encephalomyelitis
Behçet disease
Chronic lymphocytic inflammation with pontine perivascular enhancement responsive to steroids
Clinically isolated syndromes suggestive of multiple sclerosis
Myelin oligodendrocyte glycoprotein IgG-associated encephalomyelitis
Neuromyelitis optica spectrum disorder
Paraneoplastic encephalomyelopathies
Polyarteritis nodosa
Primary angiitis of the central nervous system
Sjögren syndrome
Systemic lupus erythematosus
Infectious disease
Human immunodeficiency virus
Lyme neuroborreliosis
Neurosyphilis
Progressive multifocal leukoencephalopathy
HTLV-1-associated myelopathy/Tropical spastic paraparesis
Genetic disease
Cerebral autosomal dominant arteriopathy with subcortical infarcts and leukoencephalopathy
Granulomatous disease
Lymphomatoid granulomatosis
Sarcoidosis
Granulomatosis with polyangiitis (Wegener's)
Disease of myelin
Adrenoleukodystrophy
Adult metachromatic leukodystrophy
Other
Arnold-Chiari malformation
Compressive spinal cord lesions
Vascular malformations
Vitamin B12 deficiency
Spinocerebellar disorders

Human T-lymphotropic virus, type I; IgG: Immunoglobulin G.

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Differential Diagnosis

- Common/easy to test
  - HIV, Lyme, (other infections/immunization status if starting DMT)
  - Lupus, Sarcoid, Bechet's, Sjogren's
  - Syphilis
  - Granulomatous, Sarcoid
  - B-12
  - Lymphoma
  - Cord Compression

Obj 7

DIAGNOSTICS: DIFFERENTIAL DX

Screen for "mimics", other

- ANA (lupus)
- ESR (arteritis, other inflammatory disorders)
- Lyme Titers
- B12 Level
- ACE level (sarcoid)
- Other viral diseases, HIV
- ADEM, GBS

Obj 7

KT is a 28-year-old female with PMHX significant for acute onset vision loss at age 22 (currently resolved), intermittent numbness in her face (resolved) who presents today with CC of 'dizzy/vertigo' over the last two days. She notes the hot weather seems to be an aggravating factor.

She denies headache, trauma or other neurological symptoms. She denies fever, rash or any known infections. She denies any recent travel. Prior vision loss and facial numbness were on left side with no known eliciting or alleviating factors noted; both resolved with in 3 months of onset;

Fam Hx is non-contributory

Soc Hx is + for tobacco use; She has worked on a farm for >10 years but has not had health insurance benefits.

PE/Diagnostics

- Diminished sensation on light touch, lower R. extremity
- Difficulty with RAM on L. side
- Upgoing toe on reflex testing on left side
- MRI: 2 ring like lesion in periventricular, 1 lesion in brainstem.

Obj. 1: Recognize a 'typical' clinical presentation of MS

Obj 1

ASSESSMENT AND PLAN

What is your Assessment:

28 y/o PMHx significant for resolving acute visual loss, intermittent facial numbness (resolved) with acute onset vertigo

Assessment: Vertigo, rule out MS

DDX?

TREATMENT

1 TREATMENT OF ACUTE ATTACKS

- SHORTEN DURATION AND SEVERITY
- LONG TERM BENEFIT IS UNCLEAR

2 DISEASE MODIFYING TREATMENTS

- RECOMMENDED FOR ALL PATIENTS WITH RELAPSING REMITTING
- EVIDENCE OF REDUCTION OF DISEASE ACTIVITY AND POSSIBLE REDUCTION IN SEVERITY AND PROGRESSION TO DIABILITY
- *START AT TIME OF DIAGNOSIS

3 SYMPTOMATIC TREATMENT

- INTERVENTIONS TO REDUCE SYMPTOMS THAT AFFECT QOL
- PHARMACOLOGIC AND NON-PHARMACOLOGIC

OBJ 8: DESCRIBE DIFFERENCE BETWEEN TREATMENTS FOR ACUTE RELAPSES AND DISEASE MODIFYING THERAPY AND IDENTIFY WHEN EACH IS USED

1. TREATMENT OF ACUTE ATTACKS

HIGH DOSE CORTICOSTEROIDS

- IV (typical) or Oral
- ALWAYS IV FOR OPTIC NEURITIS
- Corticotropin Injection Gel is an alternative ($$$)
- If no response, consider plasmapheresis (plasma exchange) Gluco-corticoids accelerate recovery from acute relapses/attacks

DALFAMPRIDINE (Ampyra)

- K+ Chanel blocker. May improve walking function.
- Acute and long-term use.

2. DISEASE MODIFYING TREATMENTS

- Decreased Relapse Rate
- Slower accumulation of lesions on MRI
  - Reduce clinical relapses by 30% and decrease MRI brain lesions
- Research indicates potential improvement in progression of disease but no cure.
- Studies have flaws; Need better long-term outcomes
  - Most participants have been white men of European descent.
- No single consensus on approach to DMT.
  - Discuss with patient to determine preferred approach.
  - Many new drugs receiving approval.
- EXPENSIVE, Side Effects

• …prognostic studies are often limited because few have examined all factors concomitantly in representative populations (eg, Black patients have often been underrepresented in these cohorts, which tend to enroll predominantly White patients from Europe).

2. DISEASE MODIFYING TREATMENTS (DMT)

1 Efficacy Approach: Monoclonal Antibodies

- Natalizumab, ocrelizumab, rituximab, ofatumumab, alemtuzumab.

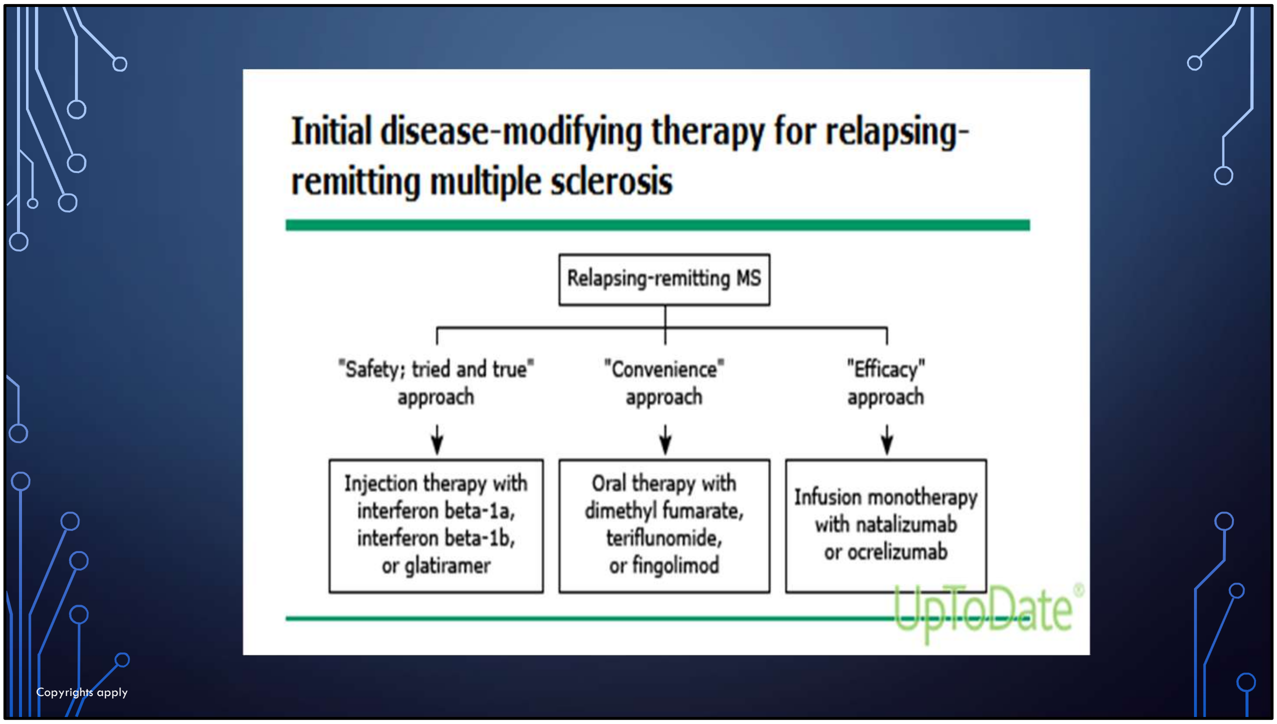
2 Convenience Approach: Oral Therapies:

- Fumarates
- Sphingosine 1-phosphate receptor modulators (fingolimod, siponimod, ozanimod, ponesimod) teriflunomide, cladribine)

3 Safety Approach: Platform Injection Therapies:

- Recombinant human interferon beta 1a and beta 1b, glatiramer acetate.

- Recommendations include patient informed decision making based on efficacy, convenience and safety. This may be an escalation (start slow and safe) vs induction approach. (more effective, more risk)


Initial disease-modifying therapy for relapsing-remitting multiple sclerosis
Relapsing-remitting MS
"Safety; tried and true" approach
"Convenience" approach
"Efficacy" approach
Injection therapy with interferon beta-1a, interferon beta-1b, or glatiramer
Oral therapy with dimethyl fumarate, teriflunomide, or fingolimod
Infusion monotherapy with natalizumab or ocrelizumab
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DMT: “SAFETY; TRIED AND TRUE” APPROACH

FOR PATIENTS WHO VALUE SAFETY OVER EFFECTIVENESS AND CONVENIENCE (FDA APPROVED FOR RRMS)

INTERFERONS

- Cytokine immune modulators
- Originally approved for RRMS, can be used in CIS, secondary progressive

GLATIRAMER Cost for one year $76,000-$87,000 ($63,000)

- One mechanism of action is to compete with myelin antigens for presentation to T cells

Side effects include flu-like symptoms, injection site relations, allergic reactions, depression and liver dysfunction. Some people develop anti-bodies to the drugs;

63

Obj 9

Interferons
=IFN beta-1a (=Avonex intramuscular injection weekly; =Rebif subcutaneous injection 2-3x/wk) and IFN beta-1b (=Betaseron: subcutaneous injection every other day)
Immune modulator; exact mechanism not known
Side effects
Injection site reaction with high doses
Flu-like side effects respond well to NSAID naproxen
Monitor liver function
May develop neutralizing antibodies
May increase depression
Glatiramer acetate=Copaxone for RRMS (daily subcutaneous injection)
Synthetic form of myelin basic protein copolymer I
Binds to major histocompatibility complex molecules
One mechanism of action is to compete with myelin antigens for presentation to T cells
Side effects
Localized injection site reactions
Self-limited (15-20 min) acute reaction that may resemble a panic attack with

palpitations, dyspnea, flushing
Not an indication to stop treatment

Safety Approach

Name	Company	Approve	Type	Indications	Other
Avonex Interferon B-1a			interferon	RRMS	
Rebif Interferon B-1a			interferon	RRMS	
Betaseron Interferon B-1b		1993	interferon	RRMS	
Extavia Interferon B-1b			interferon	RRMS	
Copaxone glatiramer				RRMS	New generic forms available. Reduced Cost

No DMT meds have FDA approval for pregnancy; beta interferon and glatiramer *may* be safe.

B-interferons were first drugs approved for tx RRMS; They are injectable/infusion. The first 'platform' drugs used for forms of RRMS were often called the ABC's as they included avonex, bataseron and copaxone.

DMT: "Convenience" approach"

For patients who value convenience (all are ORAL agents, taken daily) over effectiveness and safety (FDA approved for RRMS)

First choice (better safety profile)

- **Dimethyl fumarate** = Tecfidera® **Cost for one year $83,000**
  - Mechanism unknown (possible nrf2 activator)

Second choices

- **Teriflunomide** = Aubagio® **Cost for one year $77,000**
  - Pyrimidine synthesis inhibitor; reduction in T- and B-cell activation, proliferation and function
  - Not if liver disease or pregnant/desiring pregnancy
- **Fingolimod** = Gilenya® **Cost for one year $99,896.10**
  - Sphingosine 1-phosphate receptor modulator, sequesters lymphocytes
  - Not in diabetics (macular edema); bradyarrhythmias also occur
  - Potential teratogen

65

Convenience Approach

Name	Company	Approve	Type	Indications	Cost
Vumerity d. Fumarate (caps-DR)	Biogen	10/2019	Oral fumarate	RRMS, SPMS CIS	
Bafiertam M-fumarate (caps-DR)	Banner Life Sciences	04/2020	Oral fumarate	RRMS, SPMS	
Mavenclad Cladribine (tabs)	EMD Serono	03/2019	Purine Anti-metabolite	RRMS, SPMS	Black box: cancer, teratogenic
Aubagio Teriflunomide (tabs)	Sanfori	9/2012	Pyrimidine synthesis inhibitor <WBC in CNS, reduce inflammation	RRMS, SPMS	Birth Defects; Remains in body up to 2 years after DC
Tecfidera D.Fumarate (caps)	Biogen	03/2013	*Possible Nrf2 activator	RRMS, SPMS	

*Polyphenols of food such as garlic, sulforaphanes in cabbage, brussels sprouts, etc., green tea, onion and DHA may also activate Nrf2.

Convenience Approach

Name	Company	Approve	Type	Indications	Other
Ponvory Poesimod (tabs)	Janssen	03/2021	S-1-P receptor modulator	RRMS SPMS	
Zeposia Ozanimod (caps)	Bristol/Myers/Squibb	03/2020	S-1-P receptor modulator	RRMS SPMS	
Mayzent Siponimod (tabs)	Novartis	03/2019	S-1-P receptor modulator	RRMS SPMS CIS	
Gilenya Fingolimod (caps)	Novartis	09/2010	S-1-P receptor modulator	RRMS SPMS Pediatric Approval	Monitor HR, baseline liver, renal, vision testing

Probable Mechanism of Action: prevent lymphocytes from entering CNS.

Nearly all MS drugs cost between 63,000-104K/year!

Natalizumab=Tysabri (monthly intravenous infusion)

- Risk of progressive multifocal leukoencephalopathy (PML)
  - Rare, fatal viral (JC virus) reactivation infection affecting white matter in immunosuppressed patients
  - Risk much higher in patients with JC virus precursor (53% of MS patients); ELISA to screen for Ab to JC virus has been developed

June 2013 http://www.medscape.com/viewarticle/806294

Table. Estimation of PML Risk per 1000 Patients According to Antibody Index by Duration of Natalizumab Exposure

Antibody Index	**1 to 24 Months**	**25 to 48 Months**	**49 to 72 Months**
≤0.9	0.1	0.3	0.4
≤1.1	0.1	0.7	0.7
≤1.3	0.1	1.0	1.2
≤1.5	0.1	1.2	1.3
>1.5	1.0	8.1	8.5

Patients had no prior immunosuppressant exposure.

Efficacy Approach

Name	Company	Approve	Type	Indications	Cost
Kesimpta Ofatumumab (injx Q mo.)	Novartis	08/2020	CD20-directed cytolytic monoclonal antibody	RRMS, SPMS	
Ocrevus Ocrelizumab (IV q 6 mo.)	Genentech	03/2017	CD20-directed cytolytic monoclonal antibody	RRMS, SPMS, PPMS	
Lemtrada Alemtuzumab (injx)	Genzyme	11/2014	CD52-directed cytolytic monoclonal antibody	RRMS, SPMS	Must be in REMS programs
Tysabri Natalizumab (IV q 4wk)	Biogen	12/2004	integrin receptor antagonist	RRMS, SPMS	*Risk for PML; Check for JC virus; Enter 'touch' program

The following drugs contain box warnings for risk of PML:

Alemtuzumab
Fingolimod
Rituximab
Ocrelizumab
Dimethyl fumarate
Dimoxirel fumarate
Cladribine
Siponimod

Most drugs have approval for forms of relapsing MS. Ocrevus also has approval for PPMS.

DMT CONSIDERATIONS:

- Monoclonal Antibodies and possibly cladribine have best efficacy
- S1P receptor modulators and fumarates had intermediate efficacy
- Teriflunomide and platform drugs had lowest efficacy.

- DMT therapy currently recommended early in course of RRMS.

- Clinicians should tell patients that the relative additional benefit of highly effective DMTs comes from comparisons of their average ability to reduce relapses and new MS lesions on MRI in large groups. However, the average benefit of a DMT may not apply to an individual patient; that person may be much more likely, or much less likely, to respond to a given agent.
- None of the DMTs constitute a cure; they are only partially effective for reducing the relapse rate, and whether all or any reduce long-term disability progression is still under active investigation.

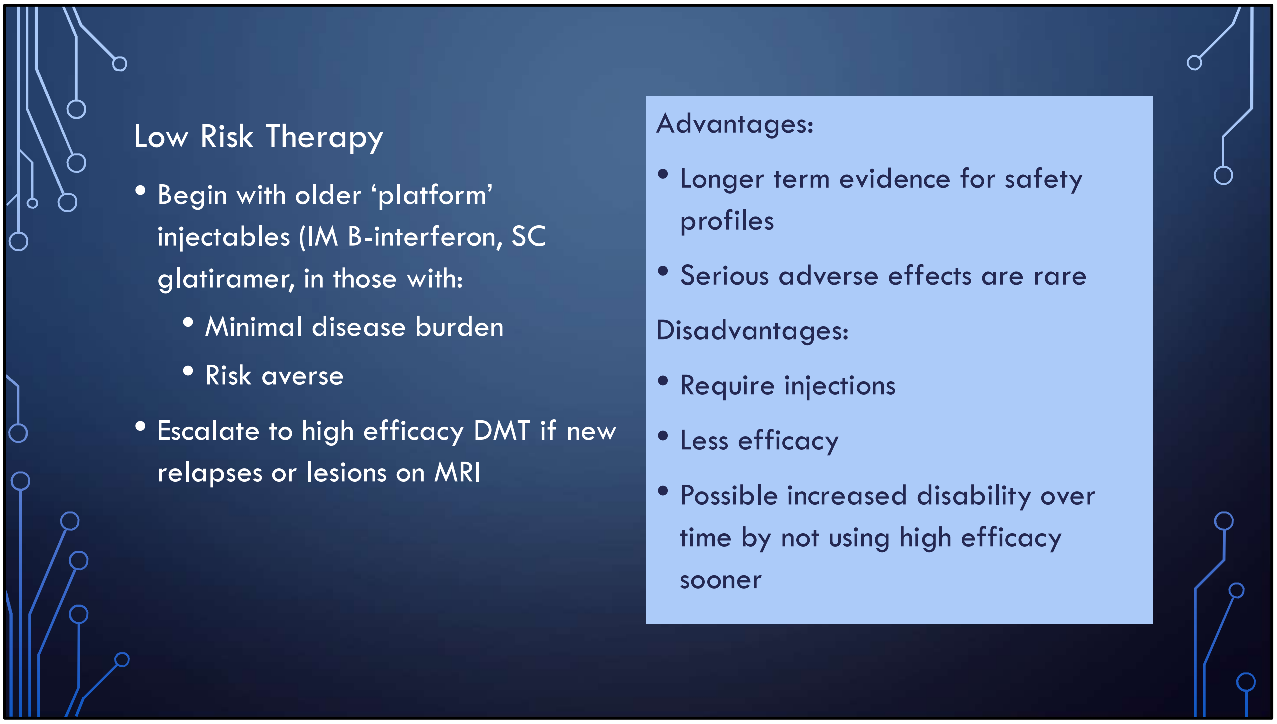
THERE IS NO UNIFORM METHOD FOR SELECTING INITIAL DMT

High Efficacy Therapy: IV natalizumab, IV ocrelizumab, SC ofatumumab.

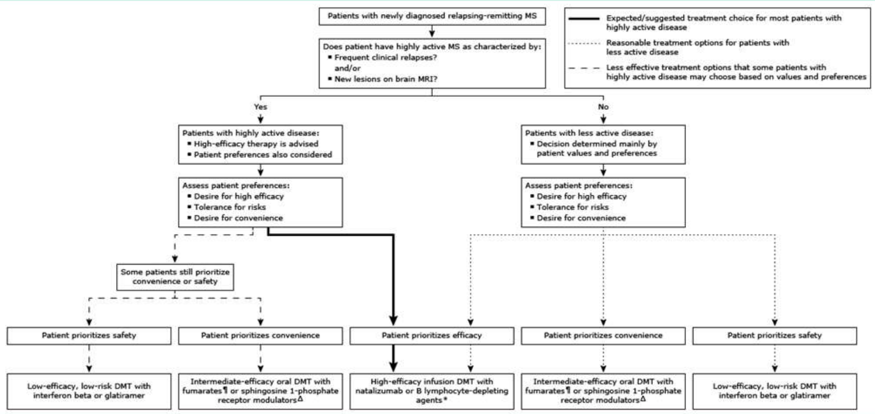
- High efficacy may help prevent relapses which tend to be more frequent at onset of MS
- Patients often younger, may be more tolerable

Disadvantages:

- Serious Infections: progressive multifocal leukoencephalopathy [PML] with natalizumab
- Risk of severe Covid infection
- Reduced efficacy of Covid vaccine
- Breakthru ds a/w natalizumab withdrawal
- Requires infusions or injections


Low Risk Therapy
Begin with older 'platform' injectables (IM B-interferon, SC glatiramer, in those with:
Minimal disease burden
Risk averse
Escalate to high efficacy DMT if new relapses or lesions on MRI
Advantages:
Longer term evidence for safety profiles
Serious adverse effects are rare
Disadvantages:
Require injections
Less efficacy
Possible increased disability over time by not using high efficacy sooner

Initial disease-modifying therapy for relapsing-remitting multiple sclerosis



There is no uniform method for selecting initial DMT for patients with relapsing-remitting MS. The following are reasonable options for choosing initial DMT:

- For patients with highly active disease and for patients who place a high value on efficacy and are relatively risk-tolerant, start with highly effective therapy using intravenous natalizumab or a B lymphocyte-depleting agent.
- For patients with less active disease and for patients who value convenience using a self-administered oral medication compared with medications requiring injections or infusions, start with an oral fumarate or sphingosine 1-phosphate receptor modulator.
- For patients who place the highest value on safety and are willing to accept lower effectiveness, start with one of the beta interferons or glatiramer. The response to DMT can be monitored by careful monitoring for acute MS attacks (relapses) and new lesions on brain MRI to ensure effectiveness for the individual patient while minimizing risk.

DMT: disease-modifying therapy; MRI: magnetic resonance imaging; MS: multiple sclerosis.
* B lymphocyte-depleting agents include ocrelizumab, rituximab, ofatumumab, and alemtuzumab.
¶ Fumarates include dimethyl fumarate, diroximel fumarate, and monomethyl fumarate.
∆ Sphingosine 1-phosphate receptor modulators include fingolimod, siponimod, and ozanimod.

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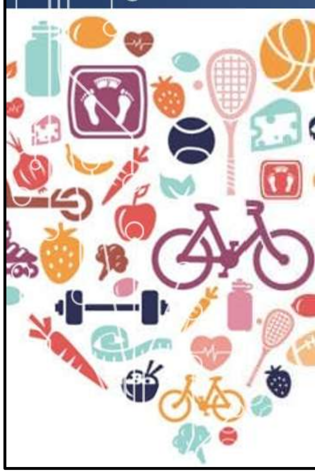
RISK OF WORSENING MS SHOULD GUIDE USE OF DMT



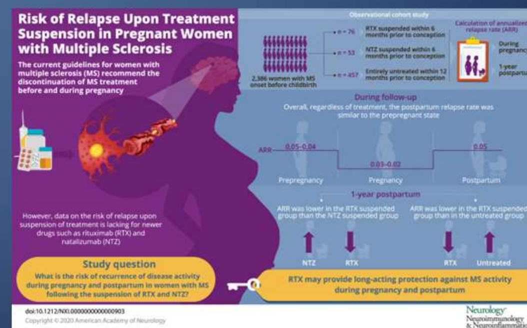
- Onset of MS after age 30, even though the risk of relapse paradoxically decreases with age
- •Motor or cerebellar onset
- •Male sex
- •Americans of Black, Asian, or Hispanic ethnicity
- Former or current smoking
- Comorbid cardiovascular disease
- Obesity
- Sedentary lifestyle
- Clinically: *poor relapse recovery, frequent early attacks, oligoclonal bands.
- Radiologic: *high burden of disease by T2 volume, ongoing disease by enhancing or new T2 lesions, *spinal cord lesions, brain atrophy.

PREGNANCY AND MS

Obj 11

- Pregnancy has a protective effect against MS relapses, but there is an increased risk of disease exacerbation in the early postpartum period.
- •Although there are no definitive data, assisted reproductive technology (ART), which includes in vitro fertilization (IVF) and other similar techniques, appears to increase the risk of MS disease activity.
- •Maternal MS may be associated with a modestly increased rate of caesarean delivery and lower infant birth weights. The choice of anesthesia for delivery in mothers with MS should be based upon obstetric concerns.
- Women anticipating pregnancy and considering DMT should have informed decision in consultation with neurologist and OB. DMT's not approved during pregnancy.

http://www.partnersmscenter.org/index.php?id=40&mn=5&sm=5-0



UptoDate recommends stopping DMT if planning a pregnancy or if pregnant; Steroids OK for acute relapses during pregnancy

http://www.steadyhealth.com/articles/Abdominal_pain_during_pregnancy__Causes___Treatments_a6.html

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- Practice Guidelines
  - Risk Category:
    - Interferons: C
    - Glatiramer: B
    - Mitoxantrone: D
    - Teriflunomide: X
      - Also detectable in semen
  - If disease stable:
    - Discontinue interferons 1-2 months before attempting pregnancy,
    - Steroids and glatiramer may be continued up the time of pregnancy
  - If disease unstable (Multiple attacks, new Gd+ lesions on MRI); attempt to stabilize disease with change in meds, adding corticosteroids.
  - If disease continues to be active, consider chemo and discuss with patient options, if male: sperm banking, if female: embryo freezing, adoption.
  - When to restart therapy:

- If breast feeding: stop breast feeding after 6 weeks and resume therapy
- If not breast feeding: restart therapy within 1-2 weeks as practicable

BASELINE ASSESSMENTS, IMMUNIZATIONS BEFORE STARTING DMT

- Update immunization status before starting DMT
- Screen for varicella, hepatitis A,B, C immunity, vaccinate if necessary.
- Vaccines should be complete at least 2-4 weeks before starting some DMT's as they may attenuate vaccine responsiveness
- Screen for JC Virus as indicated

MONITORING TREATMENT

- Follow Clinically, assess for s/s progression of disease
  - Expanded Disability Status Scale q 3 mo
- MRI at baseline/before starting DMT, 3-6 months after starting DMT, then annually.
  - May reduce annual MRI if stable
  - Consider minimizing use of gandolinium d/t accumulation in brain

3. SYMPTOM MANAGEMENT

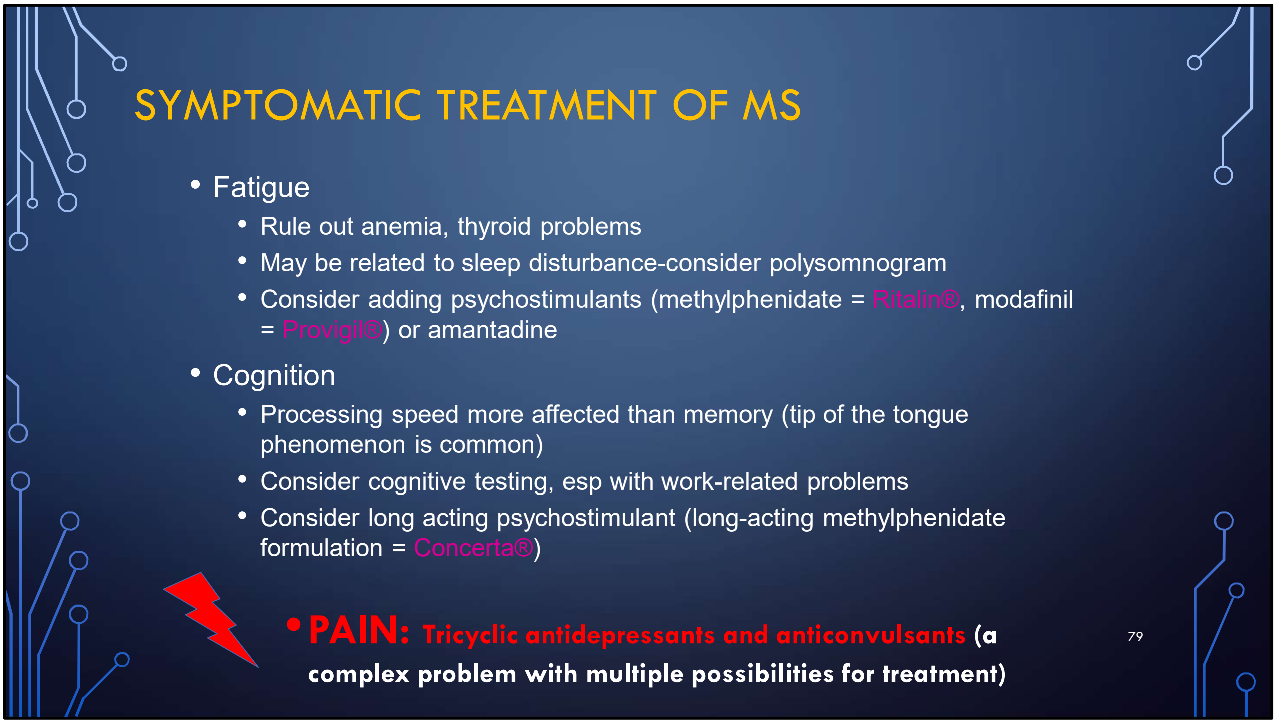
Gait impairment

- Dalfampridine = Ampyra® (extended release, oral, twice a day)
- Potassium channel blocker→possibly increasing conduction in demyelinated axons

Spasticity (muscle relaxants)

- GABA agonists **(Baclofen)** oral, or in extreme cases, via implanted intrathecal pump)
- Centrally acting α2 adrenergic agonist (Tizanidine = Zanaflex®)
- Benzodiazapine (Diazapam)
- Alert! Some patients need spasticity for walking; monitor effects!

- **Pain and paroxysmal disorders**
  - Anticonvulsants (Gabapentin = Neurontin®, carbamazepine)
  - Antidepressants (Tricyclics, SNRIs)
- **Bladder urgency** (**neurogenic bladder**)
  - **Catheterization** as needed with urology consult
  - Prevent/treat urinary tract infections



http://www.partnersmscenter.org/index.php?id=51&mn=6

Fatigue

Amantidine (Symmetrel) 100mg bid-100mg tid

Side Effects: Nausea, dizziness, insomnia

Methylphenidate (Ritalin 10-60mg in divided doses); (Concerta 18-36mg daily)

Side Effects: Hepatic dysfunction, insomnia, seizures, anorexia

Fluoxetine (Prozac) 20mg qAM-20mg bid or other Selective Serotonin Reuptake Inhibitors

Side Effects: Anxiety, anorexia, dizziness

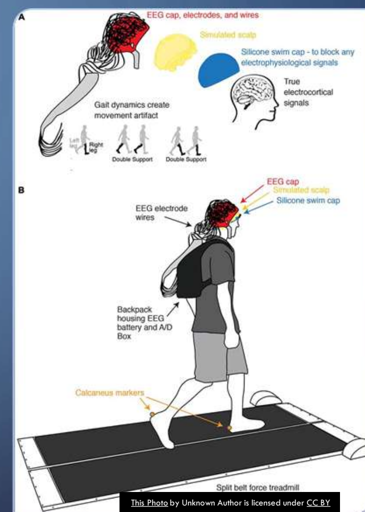
Modafinil (Provigil) 200-400mg qAM

Side Effects: Headache, nausea, rhinitis, nervousness, decreased efficacy of oral contraceptives

GAIT IMPAIRMENT

- Dalfampridine = Ampyra®   (extended release, oral, twice a day)
    - Potassium channel blocker→possibly increasing conduction in demyelinated axons

- Physical Therapy
- Guided Exercise Plan
- OMT
- Adaptive Equipment (cane, walker, shoes)
- Home Safety Evaluation
- Social Support



SPASTICITY

Gaba agonists: Baclofen

- 5mg-20mg up to QID; Intrathecal in severe cases
- Drowsy, Dizzy, Fatigue, Weakness, hypotension

Benzodiazepine: Diazepam (long acting)

- 2.5 mg-10mg up to QID
- Drowsy, Dizzy, Ataxia

Tizanidine (Zanaflex): central a-adrenergic agonist

- 2mg-12mg TID; drowsy, dry mouth, orthostatic hypotension

Dantrolene

- 25mg-100mg up to quid
- Drowsy, dizzy, weak, diarrhea, hepatic injury

- OMT
- PT
- Exercise Plan
- Relaxation Techniques/Mindfullness
- Mineral rich (magnesium) diet

PAIN: (A COMPLEX PROBLEM WITH MULTIPLE POSSIBILITIES FOR TREATMENT)

TRICYCLIC ANTIDEPRESSANTS AND ANTICONVULSANTS

Phenytoin (Dilantin) (anticonvulsant)

- 100mg bid-100mg qid; Side Effects: Nystagmus, ataxia, dysarthria, confusion, nausea, rash, hepatic injury

Carbamazepine (Tegretol) (anticonvulsant) 200mg bid-200mg qid

- Side Effects: Dizziness, drowsiness, ataxia, nausea, leukopenia, rash, hepatic injury, hyponatremia

Amitriptyline or Nortriptyline (TCA's) 10mg qhs-150mg qhs

- Dry mouth, fatigue, confusion, orthostatic hypotension, urinary retention, arrhythmia

Gabapentin (Neurontin) 100mg-600mg qid

- Dizziness, headache, drowsiness

Pregabalin (Lyrica) 50mg-100mg tid

- Side Effects: Drowsiness, dizziness, edema, dry mouth, rare rhabdomyolysis

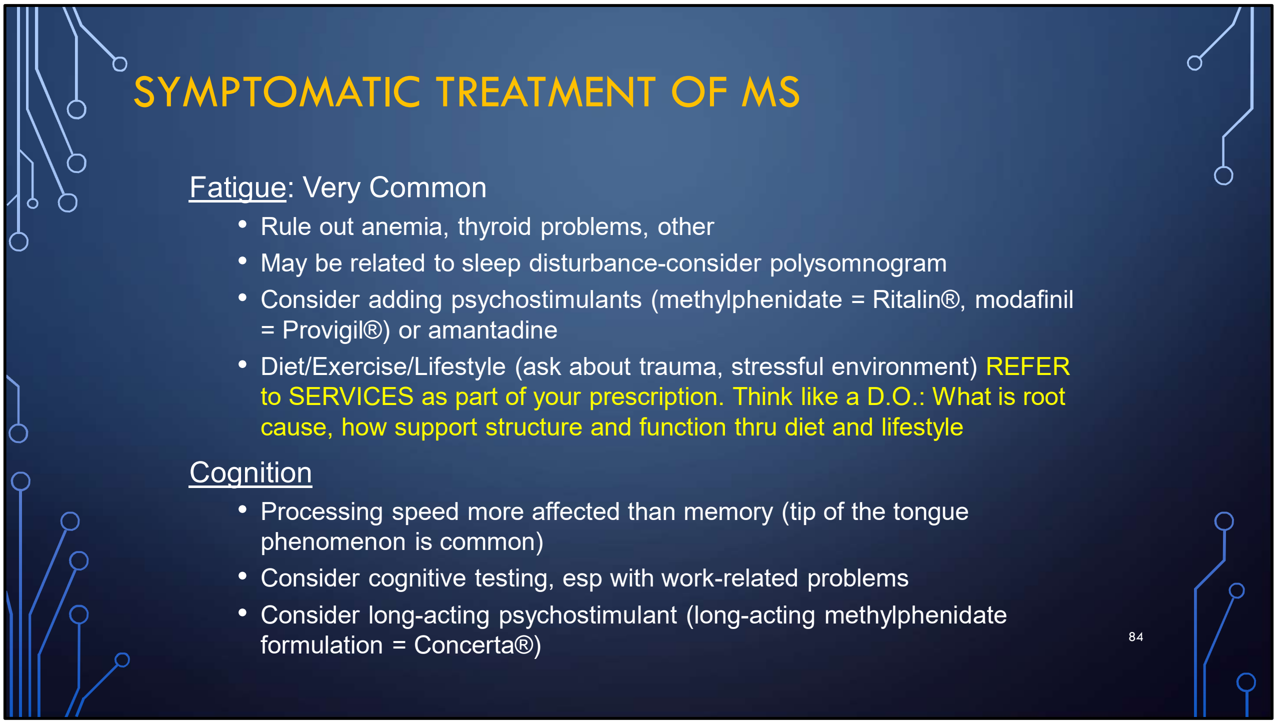
Duloxetine (Cymbalta) 30-60mg qd

- Side effects: GI side effects, blurred vision, dry mouth, muscle pain, fatigue

- Be aware of side effects and monitor
- Avoid long term opioid use

PAIN TREATMENT: PATIENT CENTERED APPROACH

- OMT
- Behavioral Health
- Physical Therapy
- Exercise Therapy
- Relaxation and Biofeedback
- Acupuncture
- Pain Management/Support Groups
- Laughter Therapy
- Nutrition



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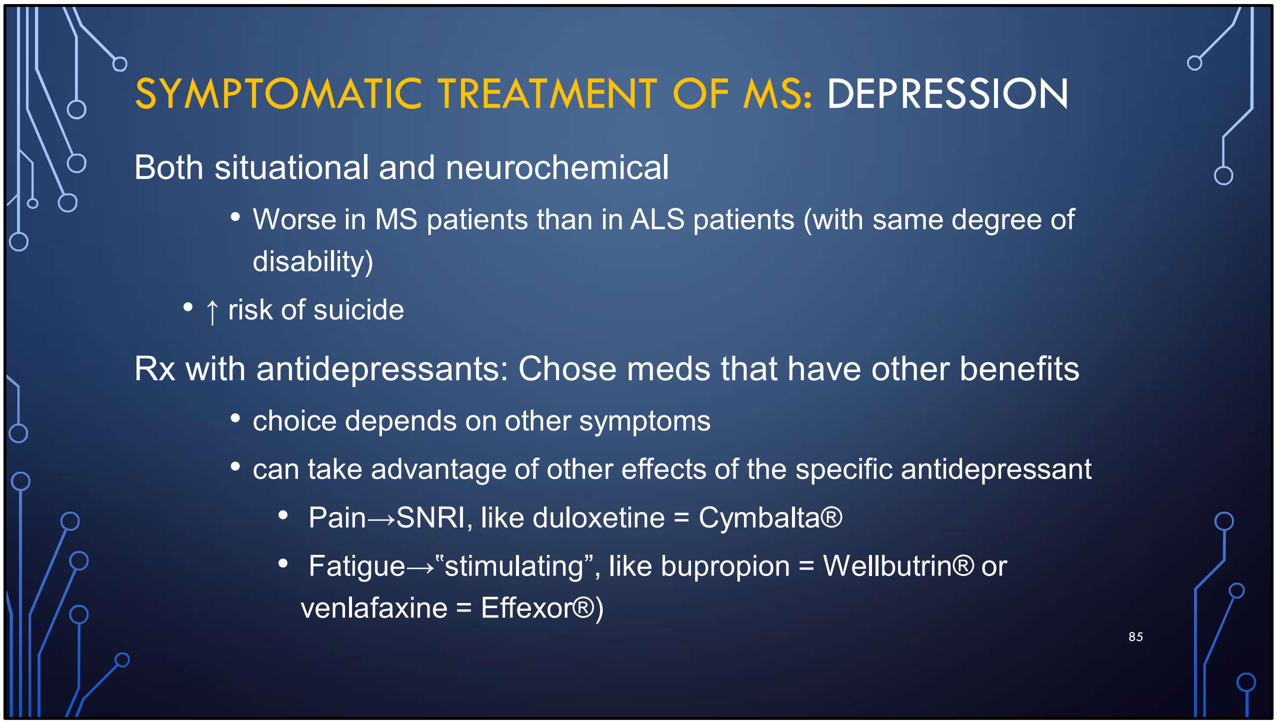

http://www.partnersmscenter.org/index.php?id=51&mn=6

Depression

Prozac (Fluoxetine) 10mg-40mg daily
Side Effects: Weight gain, sexual side effects, agitation, insomnia
Sertraline (Zoloft) 50mg-200mg daily
Side Effects: Sexual side effects, loss of appetite, GI side effects, agitation, insomnia
Escitalopram citrate (Lexapro) 10mg-20mg daily
Side Effects: Sexual side effects, nausea, diarrhea, agitation, insomnia
Citalopram (Celexa) 20mg-60mg daily
Side Effects: syncope, lightheadedness, tremor, hallucinations, sexual side effects
Venlafaxine (Effexor) 37.5mg-225mg long-acting forms given once daily
Side Effects: tachycardia, hypotension, sexual side effects, nausea, weight loss
Bupropion (Wellbutrin) 100mg-300mg in divided doses
Side Effects: lowers seizure threshold, nausea, constipation, headache,

insomnia, rare Stevens-Johnson syndrome

THINK LIKE A D.O., TREAT THE WHOLE PERSON

- OMT
- Behavioral Health
- Physical Therapy
- Exercise Therapy
- Relaxation and Biofeedback
- Acupuncture
- Pain Management/Support Groups
- Laughter Therapy
- Nutrition, Cooking Classes
- Social Work/Patient Navigators
- Pharmaceutical Interventions, DMT